

EEE554 – Probability and Random Processes

Module 5: Random Variables and Distributions

Nicolò Michelusi

Associate Professor

School of Electrical, Computer and Energy Engineering

Arizona State University

Why Random Variables?

The probability space

$$(\Omega, \mathcal{F}, P)$$

completely describes the randomness of an experiment.

However, in engineering we are rarely interested in the outcomes themselves. Instead, we are interested in numerical quantities associated with each outcome.

Examples:

- received signal amplitude
- packet delay
- queue length
- temperature
- battery lifetime

A random variable provides a numerical representation of the outcomes of a random experiment:

$$\omega \in \Omega \quad \xrightarrow{X} \quad X(\omega) \in \mathbb{R}.$$

Random Variables

A random variable is a function

$$X : \Omega \rightarrow \mathbb{R},$$

that assigns a numerical value to every outcome.

$$\omega \longmapsto X(\omega).$$

Notation

For simplicity, we usually suppress the dependence on the underlying outcome ω and write X in place of $X(\omega)$. The dependence on ω is

understood implicitly.

Example: Wireless channel experiment

Outcome ω : noise realization, fading realization, interference realization

Quantities of interest:

- $X(\omega)$ = received signal power
- $Y(\omega)$ = received SNR
- $Z(\omega)$ = packet delay

Determining $P(X > \gamma)$ may be much easier than working directly with ω .

$$A = \{ \omega : X(\omega) > \gamma \}$$

Example: Dice roll

Roll a die.

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Random variable

$$X(\omega) = \omega.$$

Outcome:

$$\omega = 4$$

corresponds to

$$X = 4.$$

Probability Distribution

$$\{\omega \in \Omega : X(\omega) \in A\}$$

Suppose we want to ask

$$P(X \in A) = P(\overbrace{X(\omega) \in A})$$

Since probability P is defined on events in the original sample space Ω , we ask:

Which outcomes ω produce values $X(\omega) \in A$?

$$(\Omega, \mathcal{F}, P)$$

These outcomes form the set

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\}.$$

Therefore,

$$P_X(A) = P(X(\omega) \in A) = P(X^{-1}(A)).$$

The random variable induces a probability measure on $\mathbb{R}$.

$$(\Omega, \mathcal{F}, P) \rightarrow (\mathbb{R}, \sigma(X), P_X)$$

Cumulative Distribution Function

The cumulative distribution function (CDF) is

$$F_X(x) = P(X \leq x).$$

Equivalent expression

$$F_X(x) = P_X((-\infty, x]).$$

Important:

Unlike a PMF (discrete) or PDF (continuous), the CDF is defined for both continuous and discrete random variables.

ACB

$A = (-\infty, x], B = (-\infty, x + \delta]$

$\lim_{x \rightarrow \infty} F_X(x) = 1$
 $\lim_{x \rightarrow -\infty} F_X(x) = 0$

non-decreasing

Properties of the CDF

$$F_X(-\infty) = 0, \quad F_X(+\infty) = 1,$$

$F_X(x)$ is nondecreasing and right-continuous.

Using the CDF

$$1 - P(X \leq x)$$

$$P(X > x) = 1 - F_X(x),$$

$$P(X \leq b) = P(X \leq a) + P(a < X \leq b)$$

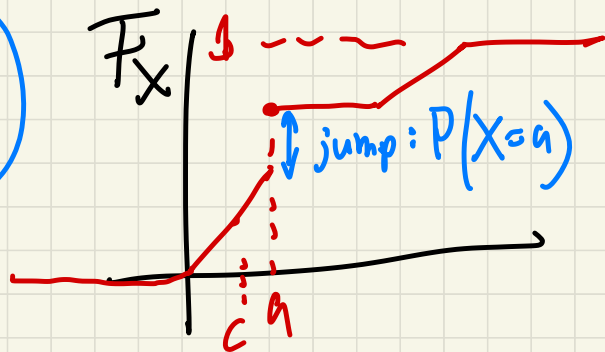
$$P(a < X \leq b) = F_X(b) - F_X(a).$$

$$P(X=a)$$

$$\{X=a\} = \{X \leq a\} / \{X < a\}$$

$$= P(X \leq a) - P(X < a)$$

$$\stackrel{!}{=} F_X(a) - \underbrace{P(X < a)}$$



$$\lim_{c \rightarrow a^-} P(X \leq c)$$

$$= F_X(a) - \lim_{c \rightarrow a^-} F_X(c) = F_X(a) - F_X(a^-)$$

Discrete vs. Continuous

The set of possible values of X ,

$$\mathcal{X} = X(\Omega) = \left\{ x = X(\omega) \mid \forall \omega \in \Omega \right\}$$

may be

- finite (die roll)
- countably infinite (number of failure before success occurs)
- uncountable (voltage value in a circuit)

Discrete Random Variables

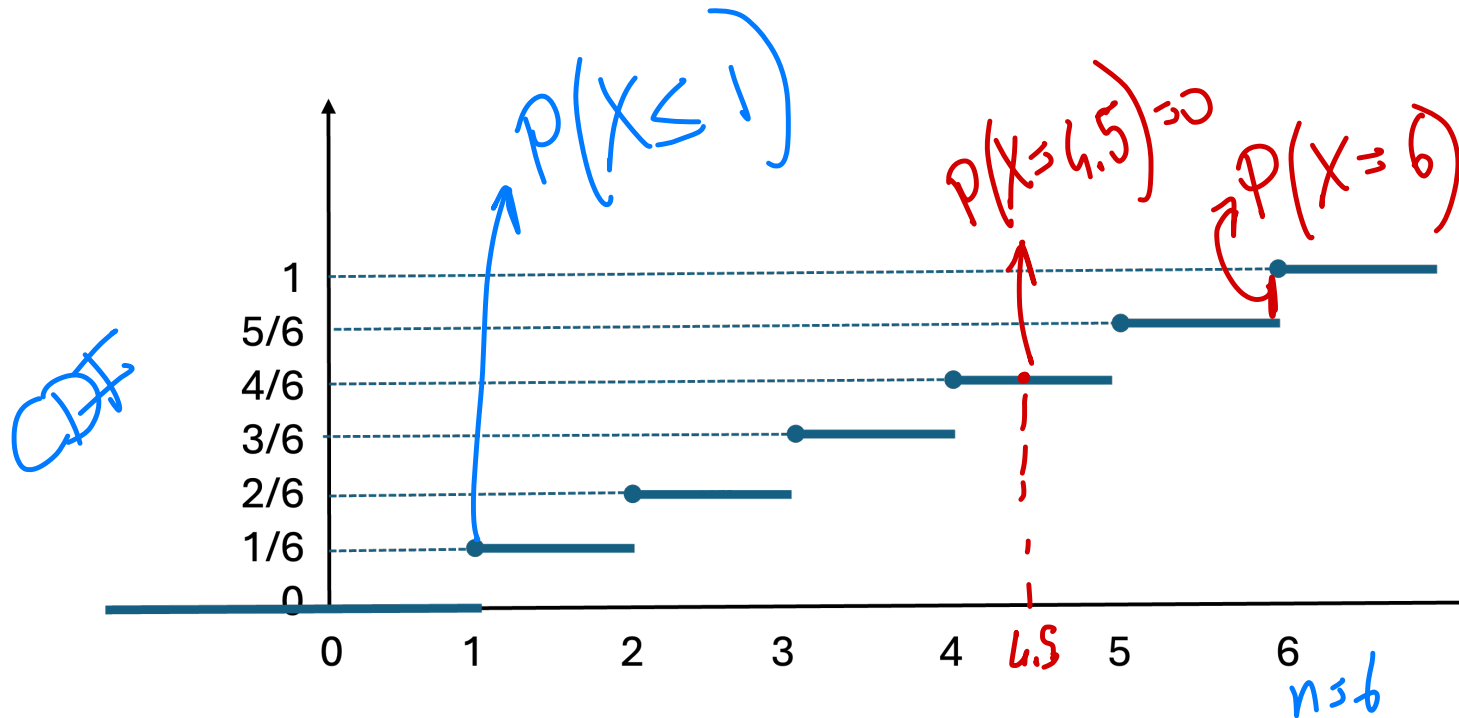
Example: dice roll

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \quad X(\omega) = \text{number shown.}$$

The CDF has a stairwise structure.

$$F_X(x) = \begin{cases} 0, & x < 1, \\ \frac{\lfloor x \rfloor}{6}, & 1 \leq x < 6, \\ 1, & x \geq 6, \end{cases}$$

$$P(X \leq a)$$



Equivalently, P_X is represented using a PMF (probability mass function):

$$p_X(x) = P(X = x) = \begin{cases} \frac{1}{6}, & x = 1, \dots, 6, \\ 0, & \text{otherwise.} \end{cases}$$

Properties

$$p_X(x) \geq 0,$$

$$\sum_x p_X(x) = 1,$$

$$P(X \in A) = \sum_{x \in A} p_X(x).$$

$$F_X(x) = \sum_{t \leq x} p_X(t)$$

$$\mathbb{P}(X \leq x)$$

Continuous Random Variables

A continuous random variable X is described by a *probability density function* (PDF)

$$f_X(x) \geq 0,$$

such that

$$P(X \in A) = \int_A f_X(x) dx.$$

In particular,

$$P(X \leq \infty) = \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

The CDF is therefore

$$F_X(x) = \int_{-\infty}^x f_X(t) dt.$$

If $F_X(x)$ is differentiable, then

$$f_X(x) = \frac{dF_X(x)}{dx}.$$

For a continuous random variable,

$$P(X = x) = 0, \quad \forall x \in \mathbb{R}.$$

Important: This does not mean that the outcome $X = x$ is impossible. Rather, individual points have zero probability; probability is assigned to intervals (and, more generally, sets) through integration.

Example: Continuous Uniform Distribution

Suppose a random number is selected uniformly from

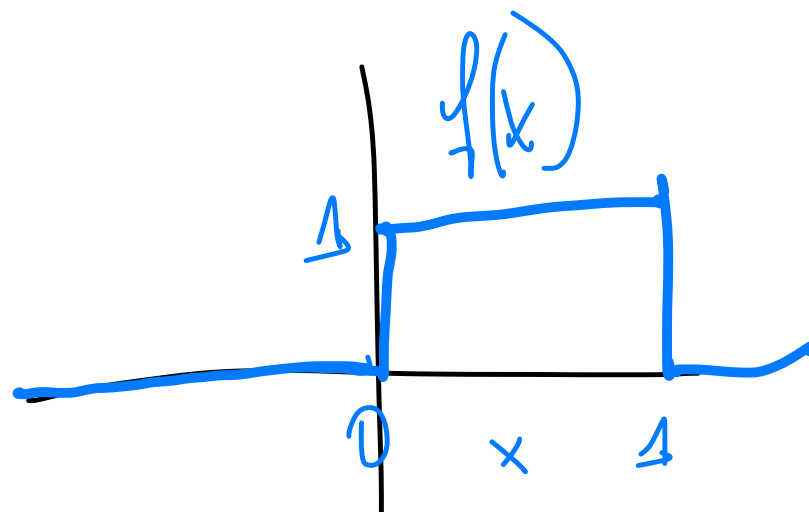
$$[0, 1].$$

Then

$$X \sim \mathcal{U}[0, 1],$$

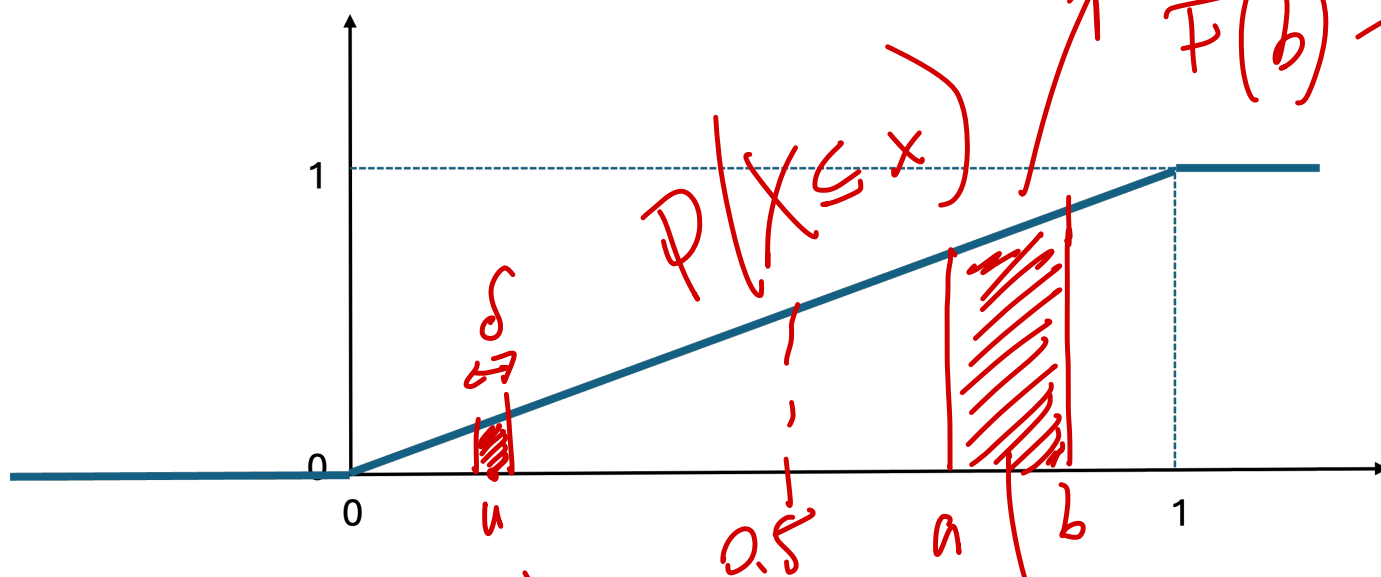
and

$$f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$



Its CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$



$$P(X=u) = \lim_{\delta \rightarrow 0} P(X \in (u-\frac{\delta}{2}, u+\frac{\delta}{2})) = 0$$

$$P(X \in (a, b]) = \text{area}$$

$$P(X \in (a/b]) = F_x(b) - F_x(a)$$

$$= \int_{-\infty}^b f(t) dt - \int_{-\infty}^a f(t) dt = \int_a^b f(t) dt$$

$$= \int_{A=(a/b]} f(t) dt$$

Discrete Uniform Distribution

Example: Outcome of a fair die roll.

Suppose $X \sim \mathcal{U}\{1, \dots, n\}$.

PMF

$$p_X(x) = \begin{cases} \frac{1}{n}, & x = 1, \dots, n, \\ 0, & \text{otherwise.} \end{cases}$$

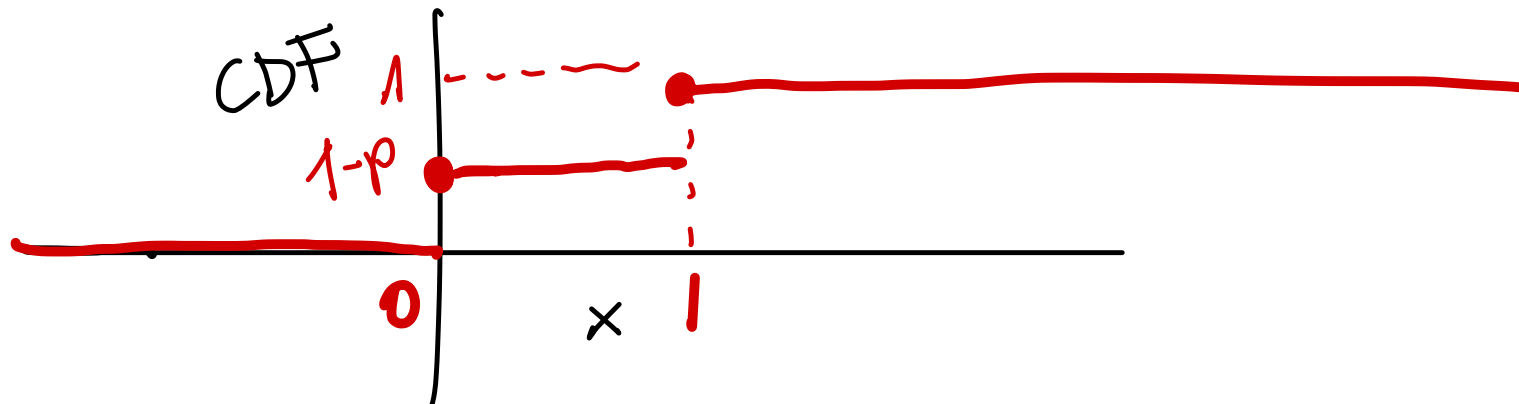
Bernoulli Distribution

Example: Success or failure of a packet transmission.

Suppose $X \sim \text{Bernoulli}(p)$.

PMF

$$p_X(x) = \begin{cases} 1 - p, & x = 0, \\ p, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$



Binomial Distribution

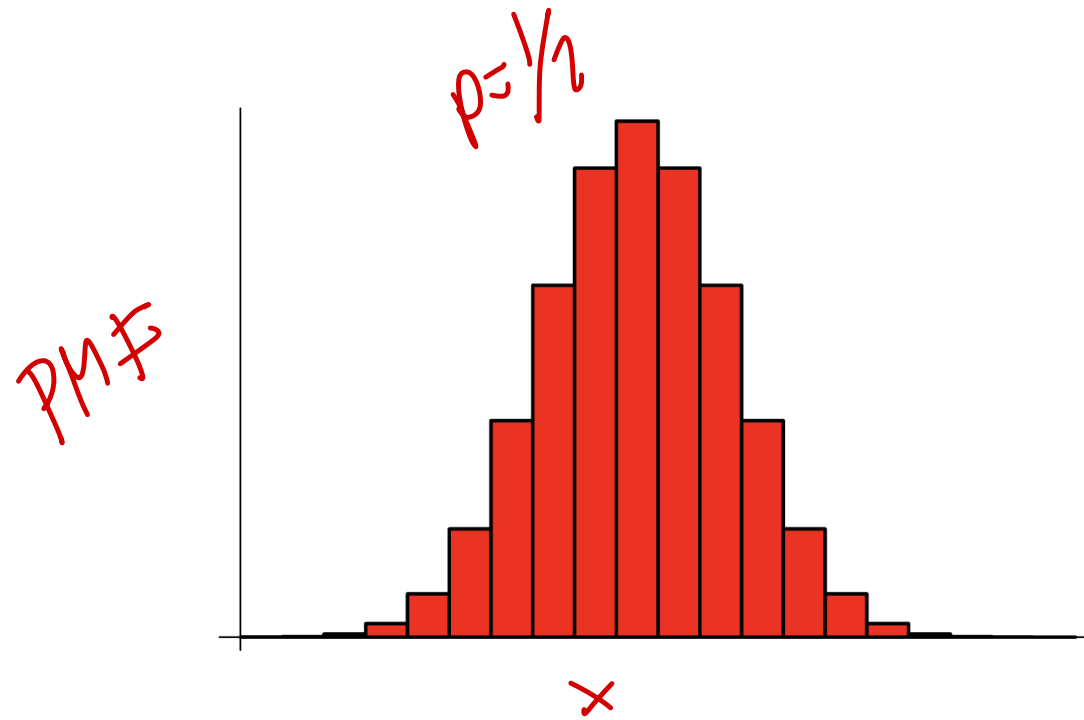
Example: Number of successful packet transmissions among n independent attempts.

Suppose $X \sim \text{Binomial}(n, p)$. Its PMF is

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n.$$

$$\frac{n!}{k! (n-k)!}$$

$$\sum_{k=0}^n p(k) = 1$$



Geometric Distribution

Example: Number of transmission attempts before the first successful packet reception.

Suppose $X \sim \text{Geometric}(p)$.

PMF

$$p_X(k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

$$\sum_{n=0}^{\infty} \alpha^n = \frac{1}{1-\alpha} \quad |\alpha| < 1$$

$$\begin{aligned} \sum_{k=1}^{\infty} p(k) &= \sum_{k=1}^{\infty} p(1-p)^{k-1} = p \sum_{n=0}^{\infty} (1-p)^n \\ &= p \cdot \frac{1}{1-(1-p)} = 1 \end{aligned}$$

Poisson Distribution

Example: Number of packet arrivals during one second.

Suppose $X \sim \text{Poisson}(\lambda)$.

$$= \lim_{n \rightarrow \infty} \text{Binomial}(n, p) \quad \begin{matrix} \lambda = n \cdot p \\ p = \frac{\lambda}{n} \end{matrix}$$

PMF

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, \dots$$

$$\begin{aligned} \sum_{k=0}^{\infty} p(k) &= \sum_{k=0}^{\infty} e^{-\lambda} \cdot \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \cdot e^{\lambda} = 1 \end{aligned}$$

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

Continuous Uniform Distribution

Example: The phase of a carrier is equally likely to lie anywhere in $[0, 2\pi)$.

Suppose $X \sim \mathcal{U}[a, b]$.

PDF

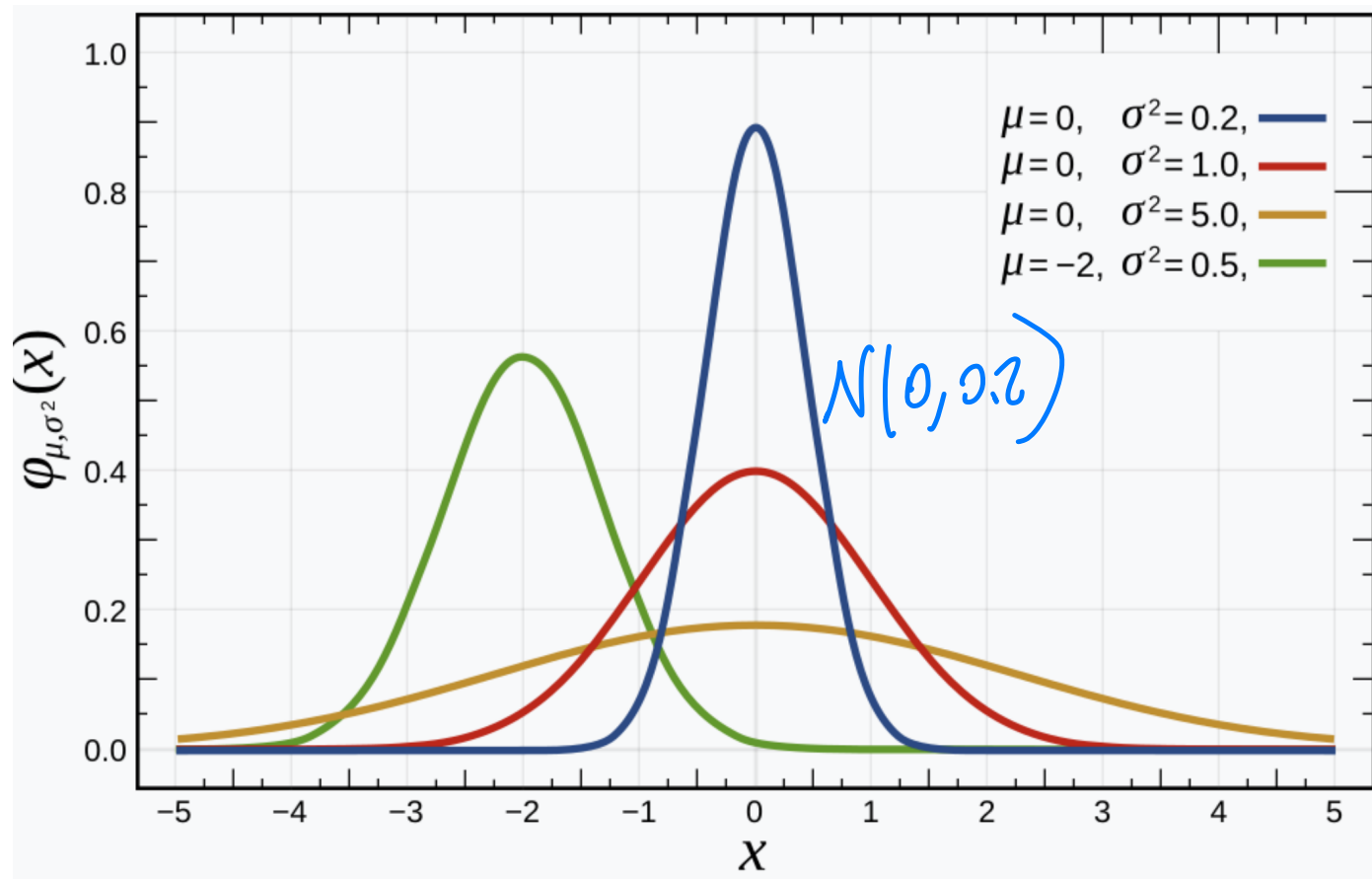
$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Gaussian Distribution

Example: Thermal noise in an electronic receiver is commonly modeled as a Gaussian random variable.

Suppose $X \sim \mathcal{N}(\mu, \sigma^2)$. PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$



Exponential Distribution

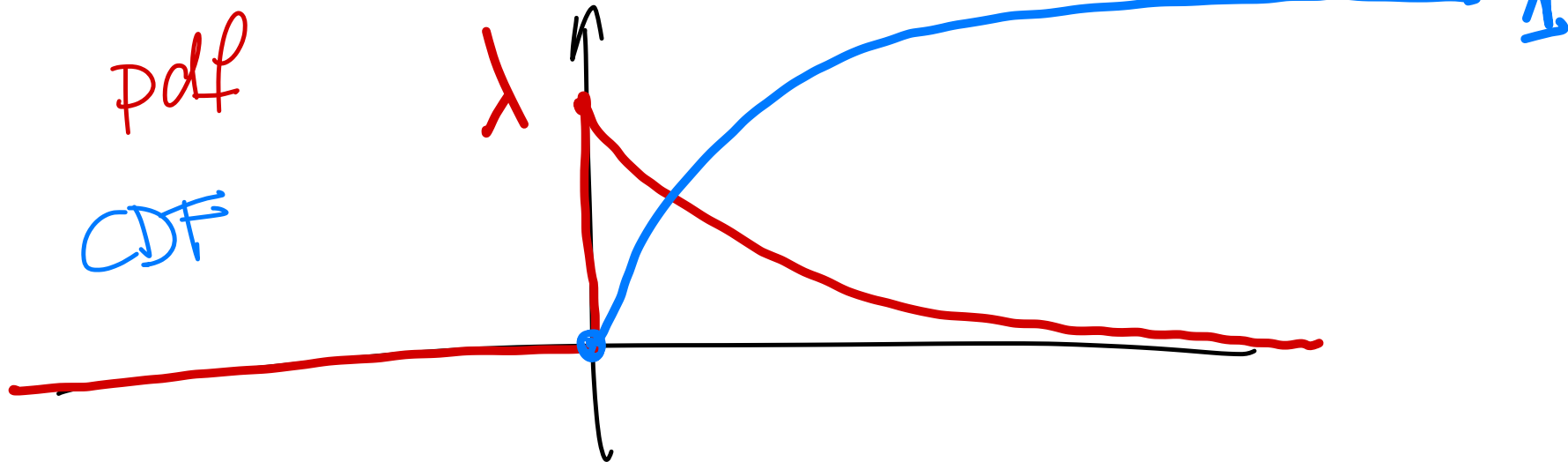
Example: Time between consecutive packet arrivals.

Suppose $X \sim \text{Exp}(\lambda)$.

PDF

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

$$CDF(x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x} \quad x > 0$$



Rayleigh Distribution

Example: Envelope (magnitude) of a wireless fading channel.

Suppose $X \sim \text{Rayleigh}(\sigma)$.

PDF

$$f_X(x) = \frac{x}{\sigma^2} e^{-x^2/(2\sigma^2)}, \quad x \geq 0.$$

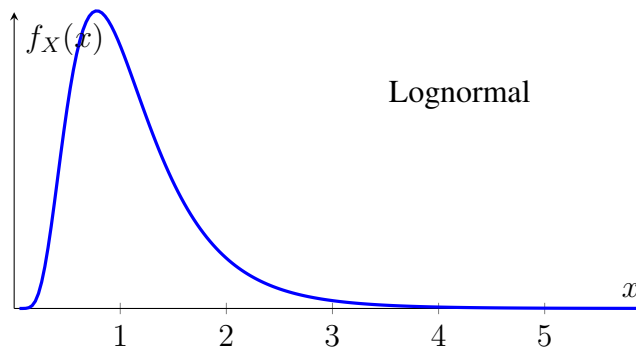
Lognormal Distribution

Example: Large-scale shadow fading in wireless communications.

Suppose $X \sim \text{Lognormal}(\mu, \sigma^2)$, i.e., $\ln X \sim \mathcal{N}(\mu, \sigma^2)$.

PDF

$$f_X(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0.$$



... and many other!

A Final Remark

Consider the probability space

$$(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), \text{Uniform}).$$

Suppose we wish to construct a random variable with CDF $F_X(x)$.

Define

$$X(\omega) = F_X^{-1}(\omega), \quad \omega \in [0, 1].$$

↓
matlab: rand

Then

$$\begin{aligned} P(X \leq x) &= P(F_X^{-1}(\omega) \leq x) \\ &= P(\omega \leq F_X(x)) \\ &= F_X(x). \end{aligned}$$

Hence, X has CDF F_X .

Simulation Recipe

Consider $\Omega = [0, 1]$ equipped with the uniform probability measure.

To generate a random variable with arbitrary CDF F_X :

1) Generate an outcome

$$\omega \in [0, 1] \quad = \text{runcmd}$$

according to the uniform probability measure.

2) Compute

$$X(\omega) = F_X^{-1}(\omega).$$

Examples

Uniform $U(0,1)$

$$X(\omega) = \omega.$$

Exponential

$$F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0.$$

Hence,

$$X(\omega) = -\frac{1}{\lambda} \ln(1 - \omega).$$

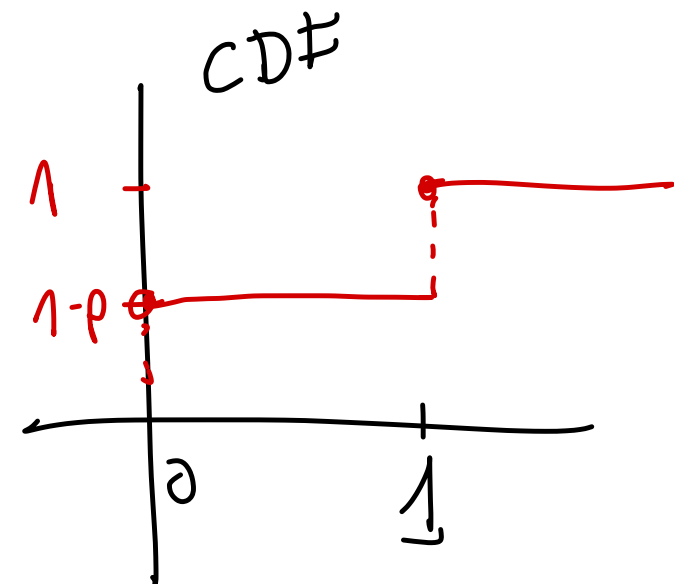
$$X(\omega) = F^{-1}(\omega)$$

$$\begin{aligned} \bullet \omega &= F(x) \\ &= 1 - e^{-\lambda x} \end{aligned}$$

$$\begin{aligned} \bullet e^{-\lambda x} &= 1 - \omega \\ -\lambda x &= \ln(1 - \omega) \end{aligned}$$

Bernoulli(p)

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$



Hence,

$$X(\omega) = \begin{cases} 0, & \omega \leq 1 - p, \\ 1, & \omega > 1 - p. \end{cases}$$

Interpretation

Starting from the probability space

$$(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), \text{Uniform}),$$

many random variables encountered in engineering can be generated by deterministically transforming the same underlying source of randomness.

Thus, different random variables can be constructed on the same probability space by applying different deterministic maps to the same underlying outcome ω .

Connection with Generative AI

A related principle appears in modern generative AI:

simple source of randomness $\longrightarrow$ transformation $\longrightarrow$ complex distribution.

Generative models often start from a simple random input, such as Gaussian noise, and transform it to produce samples from a much more complicated distribution.

The transformation is typically learned from data.

Examples include normalizing flows, diffusion models, variational autoencoders (VAEs), and generative adversarial networks (GANs).